



SIMATS Online Lab

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SIMATS ENGINEERING

SIMATS Online Programming Lab  
JAVA PROGRAMMING

CONTILA SAI DEEPAK  
192311432AB

```
27 System.out.println("Enter word to replace:");
28 String oldWord = sc.nextLine();
29 System.out.println("Enter replacement word:");
30 String newWord = sc.nextLine();
31
32 String replacedText = paragraph.replaceAll("\\b" + oldWord + "\b", newWord);
33 editHistory.append("After Replacing ").append(oldWord).append(" with ").append(newWord).append("\n");
34 .append(replacedText).append("\n\n");
35
36 // 4. Reverse each sentence individually
37 String[] replacedSentences = replacedText.split("(?<[.!?])\\s+");
38 StringBuilder reversedSentences = new StringBuilder();
39
40 for (String sentence : replacedSentences) {
41     String reversed = new StringBuilder(sentence.trim()).reverse().toString();
42     reversedSentences.append(reversed).append(" ");
43 }
44
45 editHistory.append("After Reversing Each Sentence:\n");
46 .append(reversedSentences.toString().trim()).append("\n\n");
47
48 // 5. Construct the final formatted document using StringBuilder
49 StringBuilder finalDocument = new StringBuilder();
50 finalDocument.append("=== FINAL FORMATTED DOCUMENT ===\n");
51 finalDocument.append("Total Characters: ").append(charCount).append("\n");
52 finalDocument.append("Total Words: ").append(wordCount).append("\n");
53 finalDocument.append("Total Sentences: ").append(sentenceCount).append("\n");
54 finalDocument.append("Final Text:\n").append(reversedSentences.toString().trim());
55
56 // 7. Display original document, editing history, and final document
57 System.out.println("\n----- ORIGINAL DOCUMENT -----");
58 System.out.println(paragraph);
59
60 System.out.println("\n----- EDITING HISTORY -----");
61 System.out.println(editHistory.toString());
62
63 System.out.println("----- FINAL DOCUMENT -----");
64 System.out.println(finalDocument.toString());
65
66 sc.close();
67 }
68 }
```

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SEARCH

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SIMATS ENGINEERING

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JAVA PROGRAMMING

CO2-AT1-Assignment1

QUESTIONS

Q1  
Smart Document Editor using  
Spring, Stringbuilder and  
StringBuffer  
10 marks

Q2  
Hospital Token Management  
System using List interface  
10 marks

Q3  
Airport Boarding Queue  
Management using Queue  
Interface  
10 marks

Q4  
Digital Certificate Verification  
using Set Interface  
10 marks

Q5  
Smart Inventory Management  
using Map Interface  
10 marks

5/5 answered

Q2 Hospital Token Management System using List Interface 10 marks

Hospital Token Management System using List Interface

Store patient details using an appropriate List implementation.

Register new patients.

Display patients in registration order.

Update patient information using Patient ID.

Remove cancelled appointments.

Display all patients of a particular department.

Find the first and last registered patients.

Requirement: Use a suitable List implementation

Your Code

```
1 import java.util.*;
2
3 public class Patient {
4
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7
8         List<PatientRecord> patients = new ArrayList<>();
9
10        int n = sc.nextInt(); // number of initial registrations
11        for (int i = 0; i < n; i++) {
12            int id = sc.nextInt();
13            String name = sc.next();
14            String dept = sc.next();
15            patients.add(new PatientRecord(id, name, dept, false));
16        }
17
18        int q = sc.nextInt(); // number of operations
19        for (int i = 0; i < q; i++) {
20            String op = sc.next(); // operation type (REGISTER, UPDATE, REMOVE, DISPLAY, FIND, etc.)
21
22            if (op.equals("REGISTER")) {
```

Input

```
3
101 ravi cardiology
102 sneha Neurology
103 Arun cardiology
```

Output

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```
24 String name = sc.next();
25 String dept = sc.next();
26 patients.add(new PatientRecord(id, name, dept, false));
27
28 } else if (op.equalsIgnoreCase("UPDATE")) {
29     int id = sc.nextInt();
30     String newName = sc.next();
31     String newDept = sc.next();
32
33     boolean updated = false;
34     for (PatientRecord pr : patients) {
35         if (pr.id == id && !pr.cancelled) {
36             pr.name = newName;
37             pr.department = newDept;
38             updated = true;
39             break;
40         }
41     }
42     // If not found, we simply ignore (don't print message)
43
44 } else if (op.equalsIgnoreCase("CANCEL")) {
45     int id = sc.nextInt();
46
47     for (PatientRecord pr : patients) {
48         if (pr.id == id && !pr.cancelled) {
49             pr.cancelled = true; // no remove/delete, just
50             break;
51         }
52     }
53
54 } else if (op.equalsIgnoreCase("DEPT")) {
55     String dept = sc.next();
56     boolean found = false;
57
58     for (PatientRecord pr : patients) {
59         if (!pr.cancelled && pr.department.equalsIgnoreCase(dept)) {
60             System.out.println(pr.id + " " + pr.name + " " + dept);
61             found = true;
62         }
63     }
64     if (!found) {
65         // nothing printed (compiler-friendly). You can print a message if you want.
66     }
67
68 } else if (op.equalsIgnoreCase("FIRST")) {
69     PatientRecord first = null;
70     for (PatientRecord pr : patients) {
```

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```
67
68
69     } else if (op.equalsIgnoreCase("FIRST")) {
70         PatientRecord first = null;
71         for (PatientRecord pr : patients) {
72             if (!pr.cancelled) {
73                 first = pr;
74                 break;
75             }
76         }
77         if (first != null) {
78             System.out.println(first.id + " " + first.name + " ");
79         }
80
81     } else if (op.equalsIgnoreCase("LAST")) {
82         PatientRecord last = null;
83         for (int k = patients.size() - 1; k >= 0; k--) {
84             PatientRecord pr = patients.get(k);
85             if (!pr.cancelled) {
86                 last = pr;
87                 break;
88             }
89         }
90         if (last != null) {
91             System.out.println(last.id + " " + last.name + " ");
92         }
93
94     } else if (op.equalsIgnoreCase("ALL")) {
95         for (PatientRecord pr : patients) {
96             if (!pr.cancelled) {
97                 System.out.println(pr.id + " " + pr.name + " ");
98             }
99         }
100     }
101
102     sc.close();
103 }
104 }
105
106 class PatientRecord {
107     int id;
108     String name;
109     String department;
110     boolean cancelled;
111
112     PatientRecord(int id, String name, String department, boolean cancelled) {
113         this.id = id;
114         this.name = name;
115         this.department = department;
116         this.cancelled = cancelled;
117     }
118 }
```

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```
25 for (int i = 0; i < ns i++) {
26     String name = sc.next();
27     int age = sc.nextInt();
28     String type = sc.next();
29
30     if (type.equalsIgnoreCase("VIP")) {
31         // VIP priority rule (example): higher priority for older
32         // so priority = -age (older -> smaller negative -> higher priority)
33         vipQueue.add(new VipPassenger(name, age, -age));
34     } else {
35         normalQueue.add(name + "(" + age + ")");
36     }
37 }
38
39 // Display current queue (VIP + normal separately)
40 System.out.println("Current Queue Status:");
41 System.out.println("VIP Queue (priority order):");
42 if (vipQueue.isEmpty()) {
43     System.out.println("Empty");
44 } else {
45     // Printing without removing
46     for (VipPassenger v : vipQueue) {
47         System.out.println(v.name + "(" + v.age + ")");
48     }
49 }
50
51 System.out.println("Normal Queue (FIFO):");
52 if (normalQueue.isEmpty()) {
53     System.out.println("Empty");
54 } else {
55     for (String s : normalQueue) {
56         System.out.println(s);
57     }
58 }
59
60 System.out.println();
61
62 // Boarding operations
63 // Next passenger without removing them (then board VIP / Normal)
64 int operations = n;
65 for (int i = 0; i < operations; i++) {
66
67     String next = getNextPassenger(normalQueue, vipQueue);
68     System.out.println("Next passenger: " + next);
69
70     // Board passenger (without using remove/delete)
```

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```
113     return "No passengers";
114 } else {
115     return "No passengers";
116 }
117
118 static void printRemaining(Queue<String> normalQueue, PriorityQueue<VipPassenger> vipQueue) {
119     System.out.println("VIP Remaining:");
120     if (vipQueue.isEmpty()) {
121         System.out.println("Empty");
122     } else {
123         for (VipPassenger v : vipQueue) {
124             System.out.println("VIP " + v.name + " [" + v.age + "]");
125         }
126     }
127     System.out.println("Normal Remaining:");
128     if (normalQueue.isEmpty()) {
129         System.out.println("Empty");
130     } else {
131         for (String s : normalQueue) {
132             System.out.println("NORMAL " + s);
133         }
134     }
135 }
136
137 // VIP passenger model
138 static class VipPassenger implements Comparable<VipPassenger> {
139     String name;
140     int age;
141     int priority; // smaller means higher priority
142
143     VipPassenger(String name, int age, int priority) {
144         this.name = name;
145         this.age = age;
146         this.priority = priority;
147     }
148
149     @Override
150     public int compareTo(VipPassenger other) {
151         if (this.priority != other.priority) return Integer.compare(this.priority, other.priority);
152         // tie-breaker: older first
153         return Integer.compare(other.age, this.age);
154     }
155 }
```

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```
26 String verifyId = sc.nextLine().trim();
27
28 System.out.println("\nVerification Result:");
29 if (insertionSet.contains(verifyId)) {
30     System.out.println("Certificate ID " + verifyId + " exists");
31 } else {
32     System.out.println("Certificate ID " + verifyId + " does NOT");
33 }
34
35 System.out.println("\nCertificates in insertion order:");
36 for (String id : insertionSet) {
37     System.out.println(id);
38 }
39
40 System.out.println("\nCertificates in sorted order:");
41 for (String id : sortedSet) {
42     System.out.println(id);
43 }
44
45 System.out.println("\nTotal unique certificates issued: " + insertionSet.size());
46
47 // Remove certificates (without remove/delete method)
48 System.out.print("\nEnter number of revoked certificates: ");
49 int i = sc.nextInt();
50 sc.nextLine(); // consume newline
51
52 Set<String> revokedSet = new HashSet<>();
53 System.out.println("Enter revoked certificate IDs:");
54 for (int i = 0; i < i; i++) {
55     String rid = sc.nextLine().trim();
56     revokedSet.add(rid);
57 }
58
59 // Rebuild a new set excluding revoked IDs
60 Set<String> finalSetInsertion = new LinkedHashSet<>();
61 for (String id : insertionSet) {
62     if (!revokedSet.contains(id)) {
63         finalSetInsertion.add(id);
64     }
65 }
66
67 // Final display after "remove" rebuild set
68 System.out.println("\nAfter removing revoked certificates (rebuild set):");
69 System.out.println("Certificates in insertion order:");
70 for (String id : finalSetInsertion) {
71     System.out.println(id);
72 }
```

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Q3  
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Management using Queue  
Interface  
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Q4  
Digital Certificate Verification  
using Set interface  
10 marks

Q5  
Smart Inventory Management  
using Map interface  
10 marks

3/5 answered

05 Smart Inventory Management using Map Interface

10 marks

Smart Inventory Management using Map Interface  
Store product information using an appropriate Map implementation.  
Add new products.  
Update product quantity.  
Search a product using Product ID.  
Calculate the total inventory value.  
Display products category-wise.  
Remove discontinued products.  
Display all available products.  
Requirement: Use a suitable Map implementation.

Your Code

```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8
9         Map<Integer, Product> inventory = new HashMap<>();
10
11         for (int i = 0; i < n; i++) {
12             int id = sc.nextInt();
13             String name = sc.next();
14             String category = sc.next();
15             int quantity = sc.nextInt();
16             double price = sc.nextDouble();
17
18             inventory.put(id, new Product(id, name, category, quantity, price));
19         }
20     }
21 }
```

Input

```
5
101 Apple Electronics 50 2.5
102 Milk groceries 20 1.2
103 Laptop Electronics 20 45.0
```

Output

Windows taskbar with various application icons and system tray showing 11:50 and 06-08-2026.

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```
28 if (line.startsWith("UPDATE")) {
29     // Format: UPDATE <id> <qtyChange>
30     String[] parts = line.split("\\s");
31     int id = Integer.parseInt(parts[1]);
32     int qtyChange = Integer.parseInt(parts[2]);
33
34     Product p = inventory.get(id);
35     if (p != null) {
36         p.quantity = p.quantity + qtyChange;
37
38         // Requirement: "Remove discontinued products"
39         // Since you said no delete/remove method,
40         // we mark them as discontinued by setting quantity to 0
41         if (p.quantity <= 0) {
42             p.quantity = 0;
43         }
44     } else if (line.equals("PRINT AVAILABLE")) {
45         printAvailableProducts(inventory);
46     } else if (line.equals("PRINT CATEGORY")) {
47         displayCategoryWise(inventory);
48     } else if (line.equals("PRINT TOTAL VALUE")) {
49         double total = calculateTotalValue(inventory);
50         System.out.printf("Total Inventory Value: %.2f\n", total);
51     } else {
52         // Unknown command, ignore
53     }
54
55     COMMAND++;
56     if (COMMAND > 1000) break;
57 }
58
59 sc.close();
60 }
61
62 // Display all available products (quantity > 0)
63 static void printAvailableProducts(Map<Integer, Product> inventory) {
64     System.out.println("Available Products:");
65     for (Product p : inventory.values()) {
66         if (p.quantity > 0) {
67             System.out.println(p.id + " " + p.name + " " + p.category);
68         }
69     }
70 }
71
72 // Display products category-wise
73 static void displayCategoryWise(Map<Integer, Product> inventory) {
74     // Implementation for category-wise display
75 }
```

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1923114328D

```
21     }
22     }
23
24     System.out.println("Category-wise (Available Products Count):");
25     for (String cat : countByCategory.keySet()) {
26         System.out.println(cat + " -> " + countByCategory.get(cat));
27     }
28 }
29
30 // Calculate total inventory value = sum(quantity * price) for all products
31 static double calculateTotalValue(Map<Integer, Product> inventory) {
32     double total = 0.0;
33     for (Product p : inventory.values()) {
34         if (p.quantity > 0) {
35             total += p.quantity * p.price;
36         }
37     }
38     return total;
39 }
40
41 }
42
43 // Product class
44 class Product {
45     int id;
46     String name;
47     String category;
48     int quantity;
49     double price;
50
51     Product(int id, String name, String category, int quantity, double price) {
52         this.id = id;
53         this.name = name;
54         this.category = category;
55         this.quantity = quantity;
56         this.price = price;
57     }
58 }
59 }
```

Run Save

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